LMIRA: Load-Balanced Multi-Indicator Reputation and Authentication Security Blockchain Framework for Cross-Domain Vehicular Data Sharing

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***Abstract*—In recent years, the cross-domain Internet of Vehicles (IoV) has achieved significant advancements in providing reliable privacy protection and enhanced data security for IoV cross-domain devices, facilitated by blockchain platforms. However, due to vehicle mobility and the limited management range of Road Side Unit (RSU) domains, several challenges arise during cross-domain movement: (i) blockchain load imbalance, (ii) unreliable shared information, and (iii) inefficient authentication, which compromises vehicle privacy. To address these challenges, we propose a blockchain-based load-balanced cross-domain data-sharing framework, referred to as LMIRA. First, the framework employs edge computing to build a Transformer prediction model, which predicts the transaction arrival rate and block size for the next RSU management domain. Next, a multi-indicator vehicle cross-domain data-sharing framework is developed to more comprehensively assess the quality of shared vehicle information. Finally, an efficient and secure cross-domain authentication method is introduced, ensuring the protection of vehicle privacy and information security. We conducted extensive experiments on Hyperledger Fabric and Sumo platforms using real vehicle traffic trajectories, and built a real system using Raspberry Pi 4B. The experimental results and theoretical analysis demonstrate that our proposed framework reduces latency, increases data-sharing capacity, and offers improved applicability and feasibility. Additionally, it achieves throughput at least 3.6 times higher than existing solutions, with a latency consistently below 0.6 seconds and a transaction submission failure rate at least 43.2% lower than that of current methods.**

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I. Introduction

As THE IoV evolves, it creates favorable conditions for the automotive industry to integrate with cutting-edge technologies such as artificial intelligence, 5G and high-performance computing [1], [2]. IoV offers a comprehensive platform for interaction among intelligent vehicles, enabling mobile vehicles to communicate and collaborate with one another. Cross-domain IoV typically refers to the interaction and collaborative efforts between different functional or network domains within an IoV system. By leveraging IoV technology, collaborative control between domains can be achieved, thereby enhancing overall vehicle performance and safety. For example, bus signal priority technology, which prioritizes buses can provide sufficient priority to access for buses in terms of both space and time. Secure electric vehicle charging scheduling also necessitates ultra-lightweight privacy-preserving protocols [3]. With the rapid development of IoV and various in-vehicle services, the scope of cross-domain communication between IoV is expanding, positioning cross-domain communication as a critical application scenario for IoV. Various nature-inspired and machine learning-based approaches have also been developed to address security and efficiency challenges in wireless networks [4], [5].

In the IoV, independent certificate authorities (CAs) manage different domains, making it difficult for vehicles to assess security within foreign domains. Cross-domain data sharing faces security challenges including data leakage, uncontrollable sharing, and low access efficiency [6], making secure identity authentication and data sharing the core challenge of cross-domain communication. Software-defined networking has been proposed to manage vehicular networks efficiently, complementing blockchain-based solutions [7]. However, cross-domain issues remain severe, potentially degrading communication quality and causing blockages, especially as demand surges for applications such as cooperative driving and intelligent traffic management. Blockchain technology offers a potential solution through its decentralized and tamper-proof characteristics, with consortium blockchain enabling mutually distrusting organizations to collaboratively maintain a distributed ledger. However, different domains employ varying consensus algorithms and data structures, requiring complex coordination for cross-chain communication. Disparities in vehicle density also cause significant variations in transaction volume, leading to mismatches between block size and submission rates, which increases latency and costs. This challenge is amplified by resource-constrained devices [8], [9]. Reputation systems are vital for mitigating malicious behaviors and incentivizing trustworthy information sharing [10], yet current single-indicator models remain vulnerable to attacks. Cross-domain expansion introduces further authentication challenges, including heterogeneous CAs that impede security verification, repeated authentication that creates inefficiencies, and immutable records that require sophisticated anonymization. To safeguard systems, advanced security measures are paramount [11]. These compounded limitations demand robust integrated solutions.

Motivation: Existing cross-domain IoV systems' centralized security architectures introduce single-point failures and central node vulnerabilities, constituting core challenges [12]. Blockchain technology offers solutions via inherent decentralization and tamper-resistance. Current architectures mandate secure cross-domain authentication and trust management to restrict data access to authorized entities, after which vehicles share sensor-collected sensitive data. Crucially, uneven data traffic distribution creates disproportionate node loads, exhibiting three constraints: (i) vulnerability to eavesdropping/tampering attacks compromising privacy/integrity; (ii) traffic-imbalance-induced congestion/latency degrading service; (iii) inflexible authentication mechanisms with cumbersome certificate management yielding inefficiency and privacy risks.

To address cross-domain IoV blockchain inefficiency, inadequate reputation systems, and authentication security issues, we propose LMIRA. In summary, the contributions of this paper are as follows:

1) We propose LMIRA, a blockchain-based cross-domain vehicular framework enabling load-balanced information sharing. Unlike conventional IoV solutions, it addresses data unreliability, privacy leakage, and blockchain load imbalance.
2) To our knowledge, we are the first to propose a blockchain-based information sharing framework for cross-domain vehicular networking with multi-indicator reputation management, authentication, and load balancing. LMIRA ensures information reliability through comprehensive behavioral metrics, provides efficient privacy-preserving authentication, and maintains system stability via predictive block selection with sidechain optimization.
3) Through theoretical analysis and experimental results, we demonstrate that our proposed framework (LMIRA) exhibits superior performance and is more advantageous compared to other frameworks.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the framework of LMIRA, followed by a detailed description of the design of each key module. Section VII analyzes the experimental results, and Section VIII concludes the paper.

II. Related Work

This section provides a brief review of recent advancements in related technologies within IoV cross-domain networks and offers qualitative comparisons with key representative works. A summary of the related work discussed is provided in Table I.

A. Blockchain Technology in Cross-Domain Applications

Blockchain technology has been extensively investigated for cross-domain applications owing to its immutability and security properties. Liu et al. [13] proposed a blockchain-based reliable cross-domain authentication scheme, utilizing distributed ledger characteristics to secure heterogeneous authentication processes through a dedicated protocol. Yao et al. [19] developed a lightweight secure data-sharing framework for blockchain-enabled IIoT, enabling anonymous cross-domain data access by authorized devices. Further advancing resource-constrained IoT, Hao et al. [15] designed a lightweight consortium blockchain architecture for geographically dispersed devices, facilitating intelligent autonomous access control. Ren et al. [16] introduced a blockchain-based integrity verification mechanism for cross-domain queries in outsourced databases. Complementarily, Zhong et al. [17] streamlined storage and batch computation systems, significantly reducing multi-vehicle authentication costs. However, blockchain load imbalance remains an unexplored research dimension in existing blockchain-based cross-domain IoV frameworks. Abbasinezhad-Mood et al. [20] proposed a dual-signature blockchain-based key sharing protocol specifically for multi-domain V2V communications, aiming to reduce communication rounds and enhance cross-domain authentication efficiency.

B. Reputation Management in Cross-Domain Systems

Reputation management models leverage reputation metrics to assess user trustworthiness and inter-user trust relationships. Conventional IoV systems predominantly employ centralized trust management architectures, wherein all trust data undergo centralized storage and processing. To address cross-domain challenges, Bedi et al. [21] introduced a Cross-Domain Group Recommender System. Wu et al. [22] developed a multi-objective cross-domain collaboration framework, simultaneously optimizing collaboration benefits for requester nodes and trust values for partner nodes. Hu et al. [23] innovated a dynamic evaluation window mechanism using RNNs, integrated with reputation decay dynamics. Complementing these, Wang et al. [24] designed a primary-secondary chain architecture supporting multi-trusted domains: the primary chain records authentication/authorization data, while secondary chains store domain-specific identity registrations. Current solutions exhibit a singular focus on monolithic reputation metrics, lacking comprehensive multidimensional assessment frameworks.

C. Integration of Artificial Intelligence (AI) With Blockchain

AI and blockchain have emerged as prominent research frontiers, with their integration attracting significant scholarly attention. To address service orchestration challenges in Software-Defined Networking (SDN), Zhang et al. [25] developed an AI-blockchain hybrid solution. Concurrently, Wang et al. [26] devised a blockchain-based in-vehicle crowd sensing system for

TABLE I
COMPARISON BETWEEN LMIRA AND OTHER FRAMEWORKS

	Advantages	Limitations	Cross-domain		Blockchain Performance	
			Data sharing	Approach	Latency(s)	Throughput(Tx\s)
[13],2021	A_1,A_4	L_2,L_3,L_5	–	PKI	–	[0-81]
[14],2024	A_1,A_2	$L3$,L_4,L_5	✓	IBE	0.11	326.2
[15],2023	A_1,A_2	$L3$,L_4,L_5	✓	–	[0-4.6]	–
[16],2025	A_1	L_2,L_3,L_4,L_5	✓	PKI	×	×
[17],2024	A_1,A_2,A_5	A_3,A_4	×	–	–	–
[18],2022	A_1,A_2,A_5	A_3,A_4	✓	CL-PKC	[5-27]	×
LMIRA	A_1,A_2,A_3,A_4,A_5	–	✓	JWT/PKI	[0-0.51]	[0-825.6]

$A_1(L_1)$: The framework supports (does not support) cross-domain authentication, $A_2(L_2)$: The framework supports (does not support) anonymity, $A_3(L_3)$: The framework supports (does not support) load balancing mechanism, $A_4(L_4)$: The framework supports (does not support) node trust evaluation, $A_5(L_5)$: The framework supports (does not support) unlinkability. ✓represents that the framework solves the problem, × represents that the framework can not solve the problem, – means that the framework is not appropriate for the problem. PKI: Public Key Infrastructure, IBE: Identity-Based Encryption, CL-PKC: Certificateless Public Key Cryptography, JWT: JSON Web Token.

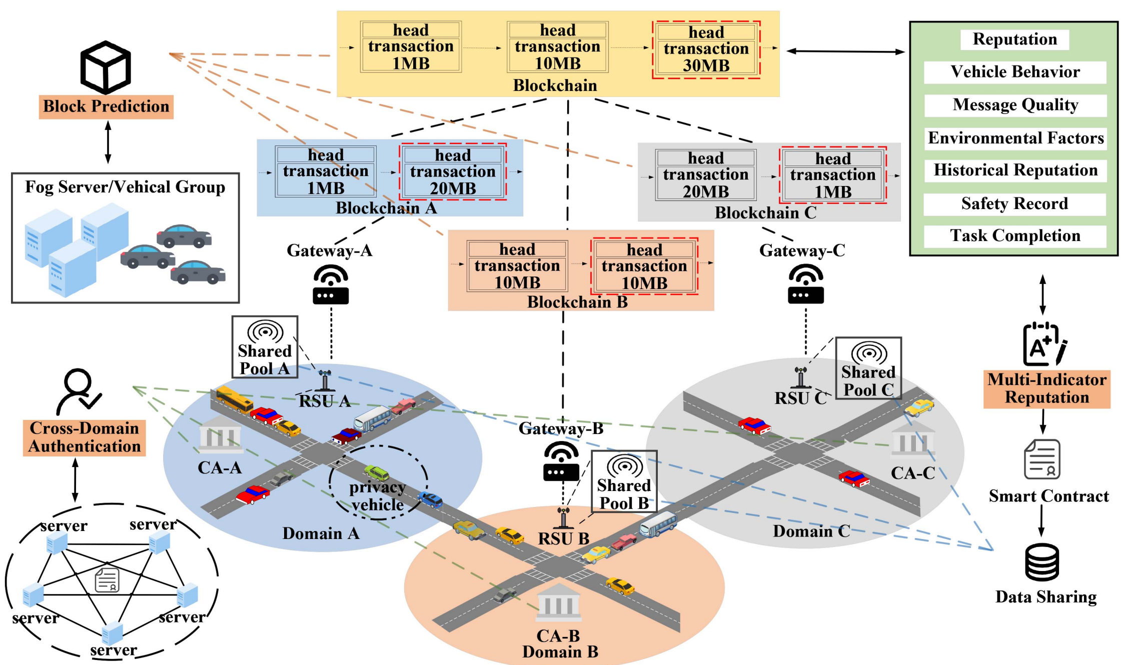

Fig. 1. System architecture of LMIRA. Comprising three modules: (1) Block Prediction: Uses scoring model to select optimal blocks, enhancing blockchain efficiency; (2) Multi-Indicator Reputation: Evaluates vehicles/RSUs comprehensively, restricting low-reputation vehicles from cross-domain access; (3) Cross-Domain Authentication: Ensures secure resource access across RSU domains.

5G IoV privacy protection, incorporating Deep Reinforcement Learning (DRL) for optimal miner and transaction selection. Boateng et al. pioneered hierarchical architectures: a multi-agent DRL framework for blockchain-authorized spectrum management [27], and a blockchain-DRL system enabling secure resource trading and autonomous slicing in 5G RAN [28]. Chen et al. [29] synthesized deep learning, blockchain, and fully homomorphic encryption to establish a Decentralized Privacy-preserving DL model for VANETs. Further advancing this domain, Alagha et al. [30] created a Blockchain-assisted multi-expert demonstration cloning framework for multi agent DRL, while Ma et al. [31] engineered an autonomous incentive-based framework for distributed controllable resources integration in grid services. Although AI has been integrated within blockchain frameworks, extant implementations neglect the performance constraints inherent to block size parameters. This limitation necessitates a novel framework capable of dynamically adjusting block dimensions through AI-driven optimization.

III. SYSTEM MODEL

This section delineates the system architecture, primary entities, and three functionally partitioned core modules of LMIRA.

A. *System Architecture*

The system architecture, shown in Fig. 1, consists of three core modules: the block prediction module, the multi-indicator reputation management module, and the cross-domain authentication module. Each domain is managed by a federated RSU cluster, which oversees information sharing pools accessible only after authentication and passing through the reputation scoring system. Of course, after appropriate cross-domain authentication, it is also possible to access information sharing pools in other domains where the CA is responsible for the authentication of each domain, as well as cross-domain authentication. Gateway to isolate the federated domains formed by individual RSUs. After RSUs collect certain relevant data, it is submitted by the gateway to the edge or mobile computing clusters consisting of Fog Service or Vehicle Group, which train the prediction model, predict the optimal block and then submit it to the main blockchain through the sidechain technology. The blockchain is responsible for recording vehicle authentication, information sharing records, and behavioral data. It also records RSU and CA authentication records at the same time to ensure the integrity of the data, and only authenticated entities can access the data on the blockchain. Through comparative analysis, consortium blockchain with permissioned access, efficient consensus, privacy, and immutability offers optimal infrastructure for multi-party IoV ecosystems, balancing security and performance.

This system employs the Raft consensus algorithm, selected for its strong consistency, millisecond-scale latency, high throughput, and implementation simplicity. These attributes render Raft the superior solution for resource-constrained IoV consortium blockchains, particularly in scenarios demanding high node trustworthiness, controlled network scale, and stringent real-time performance, where it delivers efficient consensus with moderate resource overhead.

Following the selection of a suitable block, the fog node proposes it to the leader node. Upon the Leader's validation, the block is broadcast to the fellow nodes. Subject to endorsement by at least $\frac{2}{3}$ of the fellow nodes, the leader node subsequently utilizes sidechain technology to commit the block to the chain and disseminates the result to the network. Our system consists of the following roles, as shown in Fig. 1.

1) *Gateway:* The gateway plays a crucial role in isolating the RSU federated domain to prevent uneven load distribution across the IoV communication network.
2) *RSU:* As the core component of the cross-domain management system. Each roadside unit defines a management boundary representing the maximum distance its messaging capabilities can reach. Furthermore, all associated roadside units collaborate as peer-to-peer network nodes to maintain a shared information pool. Additionally, these roadside units function as full nodes within the blockchain network, collectively participating in the consensus process to sequence, validate, and package transactions into new blocks.
3) *Vehicle:* Let T be the set of n nodes. The reputation of a vehicle $v \in T$ is denoted as R_v. Vehicles function as client entities that communicate with RSU to initiate cross-domain data sharing requests and receive shared data. Within the blockchain network, vehicles operate as lightweight nodes. After satisfying the necessary authentication and reputation criteria, they are granted access to the ledger state pertinent to their own operations, without participating in the consensus process.
4) *Fog Server/Vehicle Group:* Fog Server/Vehicle Group nodes provide edge computing services, primarily responsible for block prediction model computation and handling simple computational tasks submitted by users or official organizations. Upon completing a computational task, these nodes are rewarded with priority access to information and monetary incentives.
5) *CA:* The CA needs to authenticate each peer network node, including RSU and vehicles. It ensures the validity of digital certificates by issuing and updating them. Before each peer node joins the blockchain network, it needs to register with the CA and obtain a certificate.

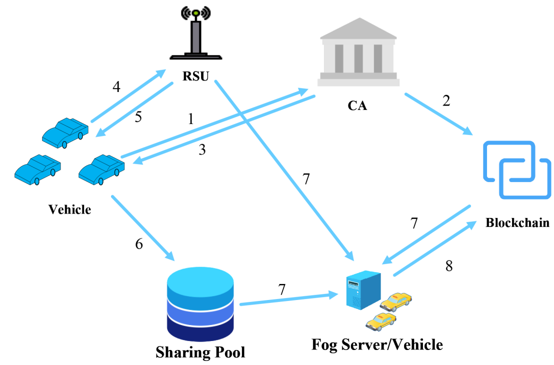

Fig. 2. General overview of LMIRA.

TABLE II
TABLE OF NOTATIONS

Symbol	Description	Symbol	Description
$\hat{D}_v^{t+1}$	Predicted domain	S_v^t	Historical block size
ID_v	Vehicle identifier	Lat_v^t, Lon_v^t	Vehicle coordinates
B^{t+1}	Predicted block	U	Block utility
$U_{perf}(b)$	Performance utility	$U_{sec}(b)$	Security utility
$C(b)$	Block cost	$R_{avg}(b)$	Average reputation
$V(b)$	Validity score	R_v	Vehicle reputation
S_{share}	Shared data size	$T_{valid}(R_v)$	Valid info proportion
B_{impact}	Reputation impact	T_i^1	Vehicle behavior
T_i^2	Message quality	T_i^3	Environmental factor
T_i^4	Historical reputation	T_i^5	Security assessment
T_i^6	Task completion	R_i	Total reputation
R_{ref}	Reputation threshold	p	Decay parameter
ω_j	Indicator weight	ϕ	Penalty factor
$Cert_i$	Digital certificate	$token_i$	Access token
$code_i$	Authorization code	PK, SK	Public/private keys
G	Block score	$G_{throughput}$	Throughput score
$G_{latency}$	Latency score	K	Fitting speed
α, β, γ	Utility weights	μ_1, μ_2	Security weights
λ_1, λ_2	Cost weights	$\omega_1, \omega_2, \omega_3$	Scoring weights
θ, ϕ	Model parameters	RSU	Road Side Unit

Fig. 2 illustrates the general overview and Table II shows the notations. The process begins with system initialization, during which the CA generates its key pairs, RSUs register with the blockchain, and smart contracts are deployed. Following initialization, the vehicle sends an authentication request to the CA (1), after which the CA records the vehicle's identity on the blockchain and generates a certificate $Cert_i$ and a token $token_i$ (2), returning them to the vehicle (3). Subsequently, the system queries the RSU for the vehicle's current reputation value (4), and the RSU returns the verified reputation (5). Once authenticated, the vehicle is permitted to access the information sharing pool (6). Concurrently, the system continuously collects vehicle trajectory data and historical blockchain information (7) to predict the optimal block and update the blockchain accordingly (Step 8). It should be noted that Steps 7 and 8 are executed periodically to ensure dynamic block optimization, while the reputation information queried in Step 4 is also updated at regular intervals to reflect the latest vehicle behavior.

Different administrative domains have independent CAs, heterogeneous credentials, and isolated reputation systems. These cause three problems: incompatible authentication, non-transferable reputation, and blockchain load imbalance. LMIRA addresses these via blockchain-based credential verification, cross-domain reputation synchronization, and dynamic block adjustment.

LMIRA adopts a consortium blockchain architecture with governance collectively formed by distributed RSU management domains. RSUs serve as core consensus nodes within the network. Within each domain, RSU clusters function as Orderers, jointly executing the consensus protocol. They order, validate, and package transactions including cross-domain data sharing requests, reputation updates, and identity authentication into new blocks, maintaining the global state of the distributed ledger. These RSUs also act as Peer nodes, executing transactions and storing full ledger copies. The CA operates as an independent identity management and access control entity, issuing digital certificates to all RSUs, vehicles, and service nodes to establish the network root of trust. Vehicles operate as clients or light nodes. To maintain system simplicity and high performance, they do not participate in consensus and only hold ledger states relevant to themselves.

The three modules collaborate: reputation dictates accessand prioritises high-quality data in prediction. RSUs fetch reputation before cross-domain to set access levels or monitoring; trajectory/behaviour analysis supports trust and threat detection, while target servers push pre-authorised codes to cut handover latency. Authentication uses JWT with anonymity.

B. Threat Model

The security considerations in this paper are determined by the adversarial abilities defined in our threat model. Based on a systematic analysis of the potential vulnerabilities in the proposed LMIRA architecture, we identify three primary attack surfaces that could compromise the system's security objectives. These attacks are widely recognized in the literature as critical threats to cross-domain IoV systems [32], [33].

- Malicious edge nodes attacks: malicious edge nodes disrupt vehicle communications by intentionally depleting local resources or interfering with communication links. Furthermore, these nodes compromise system operational integrity by submitting fabricated computational results.
- Compromised RSUs attacks: RSUs are frequently targeted by adversaries due to insufficient security mechanisms, leading to node compromise. Compromised RSUs can masquerade as legitimate infrastructure to execute spoofing attacks, inducing vehicles to establish unauthorized connections. During data collection and processing activities, these rogue nodes pose significant risks of sensitive information exposure and privacy breaches.
- Sybil attacks: attackers execute coordinated Sybil attacks within IoV ecosystems, deploying spoofed identities exhibiting attribute diversity. Through the dissemination of falsified mobility messages, they strategically manipulate vehicle routing decisions to create engineered traffic gridlock.

Based on the above threat model and system architecture, LMIRA targets comprehensive security guarantees: data confidentiality and integrity, service availability under resource depletion and traffic surges, token-based authentication and authorization, anonymous and unlinkable identity, non-repudiable on-chain logging, and resilience against malicious edge nodes, compromised RSUs, and Sybil attacks. Moreover, unauthorized entities may eavesdrop on and tamper with sensitive data transmitted through communication channels [34], [35].

C. Problem Formulation

This paper primary contribution is a blockchain-enabled cross-domain sharing framework that facilitates secure, efficient, and continuous data collection and exchange among vehicles. The targets are given as follows:

1) Block Selection Optimization: next domain prediction model can be expressed as (1)

$$\hat{D}_v^{t+1} = f_\theta(X_v^t) \tag{1}$$

Let $\hat{D}_v^{t+1}$ denote the predicted target domain for vehicle v at time $t+1$, where f_θ is the trajectory prediction model with parameters θ. The trajectory feature vector at time t is given by (2)

$$X_v^t = [ID_v, Lat_v^t, Lon_v^t, Timestamp^t] \tag{2}$$

Block prediction model model can be expressed as (3)

$$\hat{B}^{t+1} = g_\phi(X_v^t, H_{global}^t, \hat{D}_v^{t+1}) \tag{3}$$

The block generation predictor g_ϕ parameterized by ϕ outputs B^{t+1} as the anticipated block at $t+1$. We formalize the global block history feature vector as H_{global}^t, composed of the transaction arrival rate λ_v^t and historical block size S_b^t at timestep t. Based on the predictive outputs, we derive the candidate block set $\mathcal{B}^{t+1} = \{B_1, \ldots, B_k\}$. Consequently, we formulate the optimization problem as maximizing objective function (4)

Maximize:

$$U = \alpha U_{perf}(b) + \beta U_{sec}(b) - \gamma C(b) \tag{4}$$

Subject to:

$$b \in \mathcal{B}^{t+1} \tag{5}$$

$$S_{min} \leq Size(b) \leq S_{max} \tag{6}$$

$$Timestamp \geq T_{current} - \triangle T \tag{7}$$

The block utility U integrates a performance component $U_{perf}(b)$ and a security utility component $U_{sec}(b) = \mu_1 \times R_{avg}(b) + \mu_2 \times V(b)$, where $R_{avg}(b)$ denotes the average reputation value of transaction senders, $V(b)$ is the transaction validity score, and the weights satisfy $\mu_1 + \mu_2 = 1$. Concurrently, the cost function $C(b) = \lambda_1 \times C_{comp}(b) + \lambda_2 \times C_{store}(b)$ quantifies computational verification overhead and storage overhead, respectively, with $\lambda_1 + \lambda_2 = 1$.

2) Maximizing Incentive Utility: In the multi-indicator reputation module, our objective is to assess nodes from multidimensional perspectives while optimizing the incentive efficacy of the reputation system. Consequently, we formally formulate

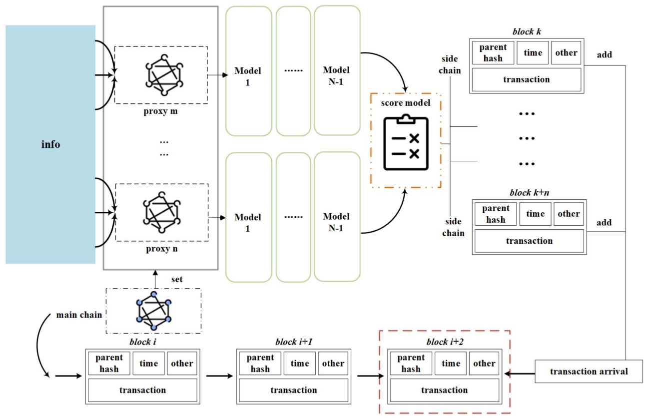

Fig. 3. LMIRA block addition process. Proxy nodes collect required information; their models process the data before evaluation by a scoring model. The optimal block is then selected and uploaded to the main chain via sidechain technology, where $block_k$ to $block_{k+n}$ denote selected blocks and $block_{i+2}$ represents the block to be generated.

the problem as follows:

$$\text{maximize} \sum_{v \in V} \Big[\kappa_1 \times S_{share}(R_v) + \kappa_2 \times T_{valid}(R_v) + \kappa_3 \times B_{impact}(R_v) \Big] \tag{8}$$

$S_{share}(R_v)$ denotes the data size of vehicle-shared messages, calculated as $S_{share}(R_v) = log(1 + DataSize)$, where DataSize represents shared message size. $T_{valid}(R_v)$ represents the proportion of verified valid information within vehicle-shared messages, defined as $T_{valid}(R_v) = \frac{N_{verified}}{N_{total}}$; here, $N_{verified}$ is the count of messages authenticated as true, and N_{total} is the total number of messages shared. $B_{impact}(R_v)$ quantifies the reputation system's impact on vehicle behavior, calculated as $B_{impact}(R_v) = \frac{N_+ - N_-}{N_+ + N_-}$, where N_+ is the number of positive behaviors, N_- is the number of negative behaviors.

IV. Dynamic Block Adjustment Method Based on Transformer

In this section, we employ a transformer model-based approach to predict vehicle trajectories and blockchain parameters. By dynamically optimizing block size and transaction upload time, we enhance the system's processing power, thereby boosting blockchain throughput and reducing latency.

A. Vehicle Trajectory Prediction and Blockchain

We deploy an agent model with predictive nodes that analyze cross-domain peers; a Transformer-based algorithm forecasts vehicle trajectories, transaction arrival rates, block sizes, and upload times. A scoring model selects optimal blocks for offloading to a side-chain, which synchronizes the new block back to the main chain once the generation threshold is met, as shown in Fig. 3. The scoring model also quantifies the impact of arrival rates and block sizes on latency and throughput, enabling dynamic block-size adjustments to optimize performance in real time.

In the prediction model setup, we utilize three models: LSTM, Transformer-decoder, and Transformer. These models process the data in two different ways. The LSTM and Transformer-decode take the same approach, using the data at the current moment to predict the data at the next moment, with the sequence step size set at 10. In contrast, the Transformer model learns from previous frames in the sequence to predict the next frame. Specifically, we use the first three frames of data in the sequence to predict the subsequent seven frames.

Conventional wisdom holds that smaller blocks reduce latency and throughput, while larger blocks increase both. In practice, however, maximizing block size to boost throughput is unsustainable. Excessive block expansion raises participation barriers for ordinary users, potentially compromising blockchain's core decentralization principle. Development must therefore prioritize maintaining decentralization, seeking equilibrium between controlled block sizing and unconstrained system performance.

Fig. 4 presents throughput and latency tests for six block sizes categorized as small (1 MB), medium (10-30 MB), and large (40-50 MB). Below 1500 transactions, small 1 MB blocks achieve roughly 20% higher throughput and 30% lower latency than larger blocks. However, beyond this threshold, 1 MB blocks saturate near 200 tps with exponentially increasing latency due to capacity constraints. Medium blocks in Fig. 4(a) show slightly lower throughput below 1500 transactions but scale effectively thereafter. Ten MB blocks peak at 767 tps around 7000 transactions, outperforming 20 MB and 30 MB blocks throughout the 1500 to 7000 range. Their latency in Fig. 4(b) follows a U-shaped curve, with minimum at approximately 2500 transactions for 10 MB blocks and 7000 for larger medium blocks; latency surges after 4000 transactions for 10 MB. Large 40-50 MB blocks exhibit throughput growth similar to medium blocks below 1500 transactions, though higher communication costs reduce throughput initially. Beyond 1500 transactions, their throughput continues scaling until saturation, with the 50 MB size sufficient for most scenarios. Rising transaction volumes congest pools and extend confirmation delays, causing latency escalation. Each block size has an optimal transaction range where performance equilibrium maximizes throughput while minimizing latency. For instance, 10 MB blocks at peak efficiency (266 tps, 0.26 s) outperform both 1 MB (215 tps, 3.01 s) and 50 MB (209 tps, 0.79 s), demonstrating the criticality of matching block size to transaction volume.

B. Predictive Scoring Model

We model the impact of the transaction arrival rate and block size on blockchain performance, as shown in Fig. 5.

We obtain different latency and throughput pairs by feeding the predicted number of transaction arrivals at the next time step along with different block sizes into the Multi-Layer Perceptron(MLP) model trained above. To determine the optimal block size for the next time step, we define a score function that

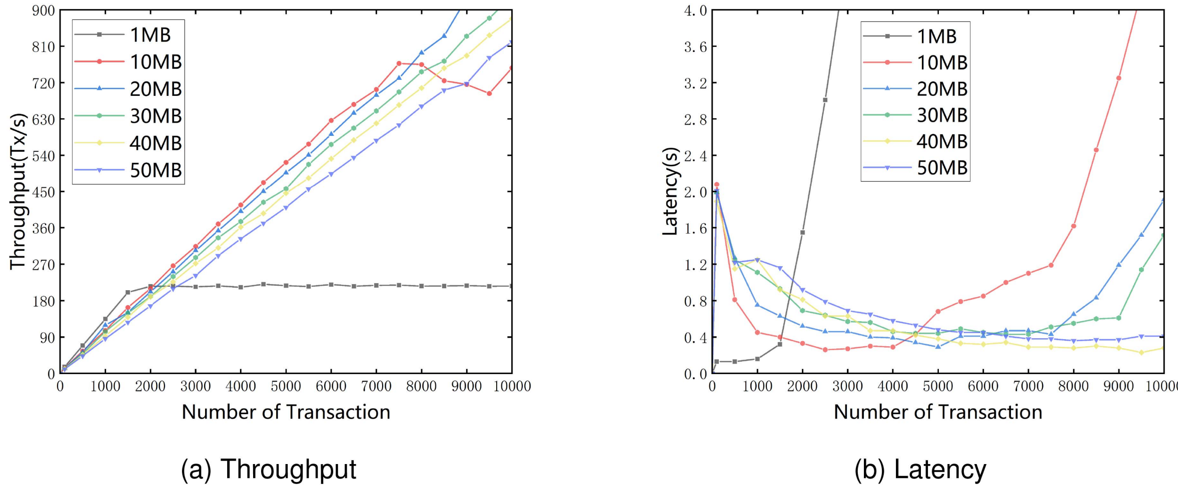

Fig. 4. Throughput and latency comparison for different block sizes. (a) Throughput comparison. (b) Latency comparison. Different block sizes exhibit an optimal range of transactions where throughput is maximized and latency is minimized.

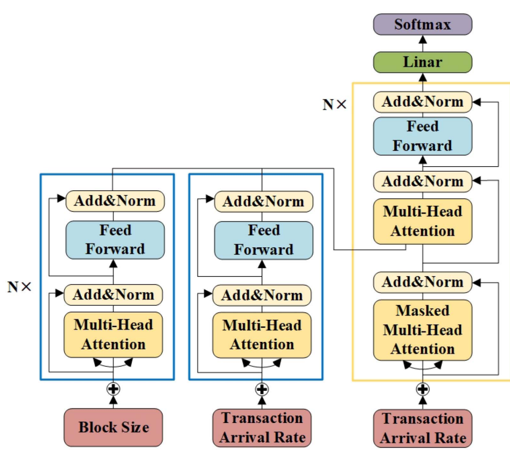

Fig. 5. Predictive model.

can be expressed as (9)

$$G = \omega_2 \times G_{throughput} + \omega_3 \times G_{latency} - \omega_1 \times K \tag{9}$$

where G represents the final score, K is the speed of model fitting, the faster the fit, the higher the value of G. $G_{throughput}$ and $G_{latency}$ are the throughput score and latency score respectively, which are weighed against whether the predicted block is the best by calculating the mean value in a blockchain-wide network. ω_1, ω_2, ω_3 are the weight coefficients, which satisfy $\omega_1 \leq \frac{\omega_2+\omega_3}{2}$ and their sum is 1. In real scenarios, these three parameters can be adjusted according to the situation. $G_{throughput}$ can be expressed as (10)

$$G_{throughput} = \sum_{i=1}^{n} \frac{p_i - p_{min}}{p_{max} - p_{min}} \tag{10}$$

where p_i represents the throughput corresponding to the block size predicted each time, and p_{max} and p_{min} denote the maximum and minimum throughputs corresponding to all blocks in the blockchain network. $G_{latency}$ can be expressed as (11)

$$G_{latency} = \sum_{i=1}^{n} \frac{l_{max} - l_i}{l_{max} - l_{min}} \tag{11}$$

where l_i represents the delay corresponding to each predicted block size, and l_{max} and l_{min} denote the maximum and minimum delays corresponding to all blocks in the blockchain network.

Getting the highest G value corresponds to the best block at the next moment, which subsequently needs to be added to the main chain, the specific process is shown in Algorithm 1. Algorithm 1 inputs a list of block sizes $S = \{S_1, S_2, .., S_n\}$ and a list of transaction arrival rates $TPS = \{Tps_1, Tps_2, .., Tps_m\}$, where S represents the sequence of block sizes that the system can accept and TPS represents the series of transaction arrival rates to be predicted, and we define $O = \{O_1, O_2, \ldots, O_M\}$ as the sequence of predicted optimal block sizes. Next, the block size and transaction arrival rate are sequentially fed into a trained MLP model. In these data, the largest G value is selected, where the size of G value depends on the throughput score and the latency score, as well as their hyperparameters(ω). After selecting the block size, the block size is subsequently adjusted to the optimal block size based on the optimal block size sequence, and the block is added to the main chain using sidechain technology.

V. Cross-Domain Sharing Scheme

In order to ensure trustworthy cross-domain data sharing for vehicles in all regions and prevent malicious vehicle nodes from sharing false information. In this paper, we establish a multi-indicator reputation model, which has the following effects on vehicle cross-domain of IoV: The vehicle's reputation can be dynamically and accurately assessed by taking into account the vehicle's behavior, messages, and environment, so as to improve the trust and reliability of the vehicle.

A. Cross-Domain Sharing Scheme

In the area of IoV, data sharing across different domains requires a trust relationship between the nodes, which will affect the efficiency and security of sharing. A reputation assessment of each vehicle and RSU involved in data sharing is necessary

Algorithm 1: Dynamic Block Adjustment.

Input: Block size list:$S = \{S_1, S_2, .., S_n\}$,Transaction arrival rate list:$TPS = \{Tps_1, Tps_2, .., Tps_m\}$
Output: Optimal sequence of blocks $O = \{O_1, O_2, \ldots, O_M\}$

```
set i = 0
while i < M do
  set G = 0, j = 0, res = 0
  while j < N do
    if S_j < S_min or S_j > S_max then
      continue
    end if
    call function MLP(S_j, Tps_i){
      get throughput and latency using Sj and Tps_i
      generate transaction Tx
    } end function
    set G_temp = ω_2 × G_throughput + ω_3 × G_latency −
      ω_1 × K
    if G_temp > G then
      set G = G_temp, res = i
    end while
    set O[i] = S[j]
end while
return O
```

to ensure their trustworthiness. In this section, the proposed cross-domain sharing model is elaborated around the setting of reputation metrics and the construction of a reputation scoring system.

Define the reputation indicator, the reputation will be determined by 6 indicators: Vehicle behavior, Message quality, Environmental factors, Historical reputation, Safety record, Task completion. These six indicators are calculated based on the parameters contained in the peer node, where the peer node $T_i (i \in \{1, 2, \ldots, n\})$, refers to the vehicles and RSUs in the same administrative domain. We define the score interval as [0,50], and the maximum assessment for each indicator is 10 points. The computation of these six metrics is given below in turn.

Indicator 1(T_i^1): Vehicle behavior, which encompasses both movement behavior and communication behavior, is evaluated for the peer node T_i^1. The calculation of T_i^1 can be expressed as (12)

$$T_i^1 = K \times N_{total} - 10 \times M_{total} \tag{12}$$

$$N_{total} = \alpha \times N_i^{in} + \beta \times N_i^{no-in} \tag{13}$$

$$M_{total} = \frac{|v_a - v_b|}{v_b} \times 100\% \tag{14}$$

where T_i^1 represents the score of indicator 1, K denotes the basic score of T_i, and N_{total} is the communication behavior score of T_i. The communication behavior score N_{total} is divided into two components: N_i^{in}, which counts the number of successful interactions between T_i and other vehicles within the domain, where α and β represent the bonus and demerit points for successful and unsuccessful interactions, respectively. Additionally, M_{total} is the communication behavior score of T_i, while v_a and v_b denote the actual speed of T_i and the domain's speed limit, respectively.

Indicator 2(T_i^2): Message quality, which refers to the authenticity and accuracy of the content of messages exchanged between T_i and other peer nodes. The message quality of peer node T_i is calculated as shown in (15)

$$T_i^2 = K \times SN_i + \sum_{j=1}^{total} M_j \tag{15}$$

$$SN_i = \frac{N_i^{in}}{N_i^{no-in}} \times 100\% \tag{16}$$

$$M_j = 1 - (e_j * f(t_j)) \tag{17}$$

$$f(t_j) = 1 + k \times (t_j - t_{normal})^2 \tag{18}$$

where T_i^2 represents the score of indicator 2, SN_i denotes the percentage of successful interactions by T_i, and N_{total} is the total number of interactions performed by T_i within the domain. Additionally, M_j represents a comprehensive evaluation of messages sent by T_i within the domain, where t_j and e_j denote the actual message delivery time and error rate, respectively. Here, t_{normal} indicates the normal or expected message delivery time, and k is an adjustment factor that controls the extent to which time deviation affects message quality.

Indicator (T_i^3): Environmental factor, which refers to the external conditions in which operates. These factors influence the efficiency of T_i in participating in the cross-domain sharing. The environmental factors of peer node T_i are quantified as shown in (19)

$$\begin{aligned} &T_i^3 = -(W_\delta \times e(\delta) + W_\varepsilon \times e(\varepsilon)) \\ &e(\delta), e(\varepsilon) \in 0, 1, 2 \end{aligned} \tag{19}$$

where T_i^3 represents the score of indicator 3, W_δ and W_ε are the weighting coefficients, satisfying $W_\delta + W_\varepsilon = 1$. Here, $e(\delta)$ denotes the road condition around T_i, where: 0 indicates excellent road conditions, 1 indicates average road conditions, and 2 indicates poor road conditions. Similarly, $e(\varepsilon)$ represents the weather condition around T_i, where: 0 corresponds to sunny weather, 1 corresponds to light rain or mist, and 2 corresponds to bad weather.

Indicator 4(T_i^4): Historical reputation behavior, which refers to the assessment of T_i reputation based on its past behavior and transaction records. We classify T_i behaviors into two categories: positive and negative. Positive behaviors include actions that enhance sharing efficiency within the region. Conversely, negative behaviors include actions that maliciously reduce sharing efficiency, such as intentionally sharing false information or violating traffic rules. The historical reputation behavior of peer node T_i is quantified as shown in (20)

$$T_i^4 = K \times \left\{ \frac{B_i^O}{B_{total}} - \phi \times \frac{B_i^N}{B_{total}} \right\} \tag{20}$$

where T_i^4 is the score of indicator 4, B_i^O is the total number of positive behaviors in T_i past, B_i^N is the total number of negative behaviors in T_i past, B_{total} is the total number of behaviors in T_i past in the domain, and ϕ is the penalty factor, the higher the number of negative behaviors, the higher the ϕ value. Where T_i^4 is the score of indicator 4, B_i^O denotes the total number of positive behaviors in T_i past, and B_i^N denotes the total number of negative behaviors in T_i past. Here, B_{total} is the total number of behaviors performed by T_i within the domain, and ϕ is the penalty factor. The value of ϕ increases with the number of negative behaviors.

Indicator 5(T_i^5): Security assessment, which refers to the mutual evaluation of T_i within the domain. The reputation of T_i is assessed based on recommendation information provided by other vehicles or infrastructures. The security assessment of peer node T_i is quantified as shown in (21)

$$T_i^5 = K \times \frac{\sum_{l=1}^{m} C_l}{m} \tag{21}$$

$$C_l = \frac{a_{T_i \to T_l}}{a_{T_i \to T_l} + b_{T_i \to T_l}} \tag{22}$$

where T_i^5 represents the score of indicator 5, m denotes the total number of security assessments of T_i within the domain. Here, C_l represents the assessment value of T_i provided by other peer nodes, $a_{T_i \to T_l}$ denotes the number of successful security assessments, and $b_{T_i \to T_l}$ denotes the number of failed security assessments.

Indicator 6(T_i^6): Task completion, which refers to the ability of vehicles participating in mobile computing to successfully complete assigned tasks. This metric is used to evaluate the reputation value of T_i. The task completion formula for peer node T_i is quantified as shown in (28)

$$T_i^6 = w_1 * complateRate + w_2 * speed + \\ w_3 * (1 - accData) \tag{23}$$

$$complateRate = \frac{complateTasks}{totalTasks} * 10 \tag{24}$$

$$speed = e^{-\lambda(responseTime - requestTime - threshold)} * 10 \tag{25}$$

$$accData = \frac{1}{n}\sum_{i=1}^{n} (processData_i - trueData_i)^2 * 10 \tag{26}$$

where T_6^i represents the score of indicator 6, w denotes the weight ratio, which satisfies $\sum w = 1$, and $complateRate$ is the task completion rate, calculated as the ratio of completed tasks($complateTasks$) to the total number of tasks($totalTasks$). Additionally, $speed$ represents the task response speed, computed from the response time($responseTime$) and request time($requestTime$). Here, $threshold$ is a queue threshold value: if the response speed exceeds the threshold, $speed$ tends to 0; otherwise, it is set to 1. The parameter λ controls the ratio of these effects. Finally, $accData$ denotes the data accuracy metric, calculated using the mean square error between the processed data($processData_i$) and the real data($trueData_i$).

Finally, these indicators are combined to derive the formula for calculating the total credibility value R_i, as expressed in (27)

$$R_i = \sum_{j=1}^{6} \left(\omega_j \times T_i^j\right)$$

$$s.t.\ 0 \leq T_i^j \leq 10, \forall j \in [1, 6] \\ \sum_{j=1}^{6} \omega_j = 1 \tag{27}$$

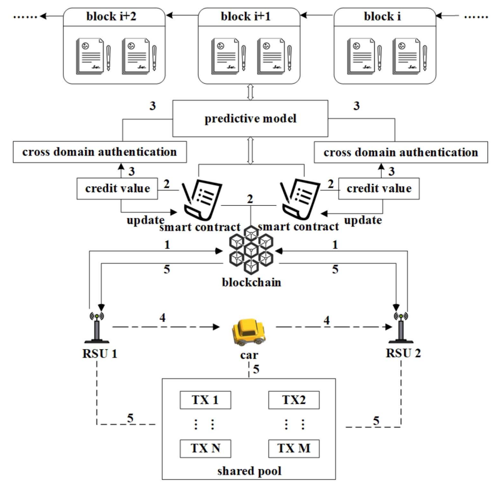

Fig. 6. Cross-domain sharing scheme.

where ω_j represents the weight of each indicator, assigned based on the relative importance of the respective indicators, and the sum of all weights satisfies $\sum \omega_j = 1$.

To mitigate selfish node behavior, a reputation decay mechanism is proposed wherein node reputation values progressively decay to a predefined threshold over time, thereby incentivizing timely participation in cooperative system activities. Reputation decay mechanism is shown in (28)

$$R_i^{t+1} = \begin{cases} \max\left(\mathrm{R_{ref}}, 10 \times \left(\frac{\sum_{i=1}^{t} \sigma(\mathrm{R_i})}{\mathrm{t}}\right)^{\mathrm{p}}\right) & R_i^t > R_{ref} \\ R_i^t & \text{otherwise} \end{cases} \tag{28}$$

Let t denote the reputation assessment time window. The reputation value of node i in discrete time t is denoted by R_i^t, with R_{ref} representing the reputation threshold. The historical reputation sequence is given by $\{R_i^1, R_i^2, \ldots, R_i^t\}$. A normalization function $\sigma(x)$ maps these values to the unit interval [0,1], while the decay parameter $p \in (0, 1)$ controls the reputation decay rate.

To establish a reputation scoring system based on pre-defined indicators, six key credibility indicators have been identified. Each indicator is assigned a specific weight to reflect its relative importance in the overall reputation assessment. The workflow of the reputation scoring system comprises five main steps: (1) collecting reputation data, (2) setting reputation thresholds, (3) implementing cross-domain authentication, (4) monitoring and updating, and (5) facilitating data sharing. The working process of the model is illustrated in Fig. 6.

1) *Collecting Reputation Data:* Gathers historical reputation values from blockchain and real-time behavioral data from vehicles/nodes. Uses hash encryption during collection/processing to protect user data.
2) *Setting Reputation Threshold:* Federation chain automatically executes smart contracts to set thresholds. Only

Algorithm 2: Intra-Domain Authentication.

Input: ID_i authentication documents
Output: Authentication results, certificate and token
if no authorization $code_i$ **and** no $token_i$ **then**
  set $C = E_{PK_{CA}^{Area1}}(documents)$
  CA decrypt $documents = D_{SK_{CA}^{Area1}}(C)$
  if ID_i **in** BlockRestrictedList **or** info is not legal **then**
    return FALSE
  else
    generate $Cert_i$ and $token_i$
    generate transaction Tx
    return $Cert_i, token_i$
  end if
else if ID_i have authorization $code_i$ from CA_{Area2} **then**
  send $code_i$ to CA_{Area1}
  CA_{Area1} **send** $message_{code}$ to CA_{Area2}
  CA_{Area2} **validate** $code_i$
  if $code_i$ is valid **then**
    CA_{Area2} **send** $message_{userInfo}$ **to** CA_{Area1}
  else
    return FALSE
  end if
  return $Cert_i, token_i$
else if $token_i$ has expired **then**
  ID_i **send** $message_{renewal}$ **to** CA_{Area1}
  if $message_{renewal}$ is valid **and** $time < time_{max}$ **then**
    modify token expiration time
    return $Cert_i, token_i$
  else
    return FALSE
  end if
end if

entities exceeding thresholds may communicate cross-domain.

3) *Implementing Cross-Domain Authentication:* Establishes mechanism permitting cross-domain communication exclusively for high-reputation vehicles/nodes.
4) *Monitoring and Updating:* RSUs track vehicle/node behavior, updating reputation scores. Cross-domain privileges are restricted for scores below threshold.
5) *Data Sharing:* The system establishes an information-sharing pool for qualified vehicles/nodes, enabling secure sharing of: vehicle status, environment sensing data, vehicle failure information, and traffic management details. All shared data is immutably recorded on the blockchain to enhance road safety, optimize traffic flow, and personalize the driving experience.

LMIRA establishes a reputation-based cross-domain data sharing framework employing multi-dimensional evaluation of vehicular trustworthiness. Its multi-criteria scoring model synthesizes vehicle behavior, message quality, environmental factors, historical reputation, safety record, and task completion metrics.

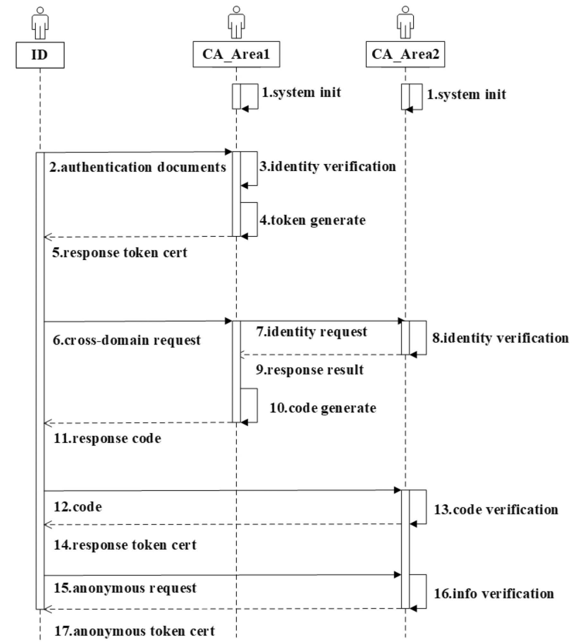

Fig. 7. Cross-domain authentication process overview: ID denotes the authentication subject, CA_{Area1} its current domain, and CA_{Area2} the target domain for cross-domain authentication.

VI. Cross-Domain Authentication Program

In this section, we detail the cross-domain authentication scheme of LMIRA, which provides a sustainable and anonymous cross-domain mechanism based on authorization codes.

Vehicle cross-domain authentication in the IoV refers to a technical solution designed to verify the identity and authority of vehicles across different trust domains or organizations. It is worth noting that the CA introduced in this framework serves as a fully trusted government-appointed agency. The cross-domain authentication scheme is divided into four phases, as illustrated in Fig. 7.

A. Intra-Domain Authentication

In the intra-domain authentication phase, LMIRA performs different actions based on the vehicle's authentication status, as outlined in Algorithm 2. We classify the authentication process into the following three cases: No authorization $code_i$ and no $token_i$. Authorization $code_i$ from CA_m^{Area2} is present. $token_i$ having expired. For this analysis, we assume the current domain is Area1, and authentication is required. We define the vehicles requiring authentication as ID_i. Each domain maintains a no-authentication list, denoted as $BlockRestrictedList = \{ID_1, ID_2, \ldots, ID_n\}$.

In the first type of authentication (no authentication $code_i$ and no $token_i$), ID_i prepares a file containing the applicant's information, such as the public key, signature algorithm, expiration date, and other relevant details. The plaintext message format is

$$\{Authenticate, ID_i, PK_i, Sign\},$$

ID_i encrypt the file using PK_{CA}^{Area1} and sends it to the CA. The CA decrypts the file using SK_{CA}^{Area1} and verifies the decrypted message to confirm the legitimacy of the identity and check whether ID_i is listed in the blacklisted $BlockRestrictedList$. If the verification is successful, the CA issues the certificate $Cert_i$ generates $token_i$, and returns them to ID_i. The authentication of ID_i is completed after verifying the legitimacy of the certificate.

In the second type of authentication (where ID_i possesses the authentication $code_i$ from CA_m^{Area2}), ID_i holds the authentication code $code_i$ for the other domain (domain Area2). The plaintext message format is

$$\{AuthenticateCode, code_i, PK_i, Sign\},$$

CA_{Area1} sends $message_{code}$ to CA_{Area2} in the following format

$$\{CrossDomainCode, code_i, PK_{CA}, Sign\},$$

CA_{Area2}, after verifying the validity of the authentication code $code_i$, returns $message_{userinfo}$ in the following format

$$\{CrossDomainInfo, Cert_i, PK_i, token_i\},$$

Finally, the $token_i$ and certificate $Cert_i$ are securely returned to ID_i.

In the third type of authentication (where $token_i$ has expired), the expiration of $token_i$ typically occurs due to ID_i inaccurate judgment of their time within the domain or exceeding the maximum allowed time for a one-time request. This approach is simpler: ID_i only needs to send a $message_{renewal}$ requesting the renewal of the token's expiry time. The plaintext message format is

$$\{TokenExpire, ID_i, Cert_i, PK_i, token_i, time\},$$

where $time$ refers to the new expiry time.

B. Cross-Domain Authentication

Assuming that the vehicle ID_i is currently in domain Area1, ID_i can actively initiate the application. Alternatively, if the vehicle trajectory prediction module predicts that ID_i will soon enter domain Area2, ID_i may actively or passively initiate cross-domain authentication for Area2 while still in Area1, as outlined in Algorithm 3. ID_i first sends $message_{license}$ to CA_{Area1} of domain Area1. The structure of $message_{license}$ is

$$\{Licensing, Area2, CA_{Area1}, Cert_{Area1}, PK_i\},$$

CA_{Area1} verifies the legitimacy of the identity information and confirms that ID_i is not listed in $BlockRestrictedList_{Area1}$. Subsequently, CA_{Area1} sends $message_{crossLicense}$ to CA_{Area2}. The plaintext message format is

$$\{CrossLicense, ID_i, info, PK_i, Sign\},$$

CA_{Area2} verifies the identity information of ID_i and checks whether it is listed in $BlockRestrictedList_{Area2}$. If the verification is successful, CA_{Area2} notifies CA_{Area1} that the information is correct. Subsequently, CA_{Area1} generates the authorization code $code_i$ securely records the details of this authentication on the blockchain, and returns the authorization code to ID_i.

Algorithm 3: Cross-Domain Authentication.

```
Input: ID_i authentication documents
Output: authorization code_i
 ID_i send message_license to CA_Area1
 if info is not valid then
   return FALSE
 else
   CA_Area1 send message_crossLicense to CA_Area2
   if ID_i in BlockRestrictedList_Area2 or info is
   not valid
     return FALSE
   else
     generate code_i
     generate transaction Tx
     return code_i
   end if
 end if
```

Algorithm 4: Identity Anonymity.

```
Input: ID_i authentication documents
Output: token_temp, Cert_temp
 ID_i send message_anonymity to CA_Area1
 if ID_i in BlockRestrictedList_Area1 then
   return FALSE
 else
   generate token_temp and Cert_temp
   set hashmap[token_temp] = token_i
   set hashmap[Cert_temp] = Cert_i
   generate transaction Tx
   return token_temp, Cert_temp
 end if
```

C. Identity Anonymity

If domain Area1 has already completed authentication and ID_i has obtained a token and a digital certificate, ID_i can request an anonymous identity to ensure that its behavior records remain inaccessible to other users, as outlined in Algorithm 4. ID_i sends an identity anonymity request($message_{anonymous}$) to domain Area1 in the following format

$$\{Anonymous, token_i, Sign\},$$

CA_{Area1} verifies the validity of $token_i$ and checks whether ID_i is listed in $BlockRestrictedList_{Area1}$. If the authentication is successful, CA_{Area1} generates temporary credentials, $token_{temp}$ and $Cert_{temp}$, which are linked to the original $token_i$ and $Cert_i$ using an internal hashmap in Area1. This linkage ensures accountability and prevents ID_i from evading responsibility in case of inappropriate behavior.

Intra-domain authentication occurs upon vehicle arrival, involving only the local CA to issue credentials. Cross-domain authentication is proactively triggered before entry, either voluntarily or via trajectory prediction, engaging both CAs to pre-issue an authorization code. Upon arrival, this code streamlines the local procedure, reducing handover latency; that is, proactive pre-authorization expedites reactive verification.

Authentication is implemented via dual-layer verification: the target RSU first validates the cryptographic signature of the authorization code, then queries the blockchain ledger to confirm CA registration, vehicle blocklist status, and reputation score. Cross-domain authentication is proactively initiated before handover using trajectory prediction, allowing both CAs to pre-issue codes that streamline entry. For anonymity, the CA issues session-specific ephemeral credentials that prevent external linkability, while maintaining an encrypted internal mapping for traceability if malicious behavior is detected, thus offering accountable anonymity.

D. Information Access

Once ID_i has obtained either real or anonymous identity information, it can use $token_i$ to request access to content when needed. The CA verifies the validity of $token_i$, and this information can be securely submitted to the CA through encryption using PK_{CA}^{Area1}. Additionally, to ensure the confidentiality of $token_i$, a zero-knowledge proof can be employed to demonstrate that ID_i possesses the $token_i$. Furthermore, all access records of ID_i are stored on the blockchain. CA_{Area1} periodically checks the number of violations by ID_i.

The channel system facilitates the creation of a private subnet within the blockchain network, enabling vehicles, RSUs, and CAs to join as members. This configuration ensures a secure authentication process. Members of a channel can only access transaction information within their own channel and are unable to view the contents of other channels, effectively isolating and protecting data. Transactions within a channel are confirmed and executed exclusively by the nodes of that channel, eliminating the need for participation from nodes across the entire network. This approach reduces network overhead and latency, thereby enhancing authentication efficiency.

VII. Performance Analysis

This section describes the formal simulation setup and presents the results of the LMIRA framework.

A. Simulation Setup

We deployed Hyperledger Fabric 2.3 on a MacBook Pro equipped with a 64-bit Apple M1 Pro chip and 16 GB RAM. Additionally, we set up a Linux virtual machine (VM) on the same MacBook Pro, configured with 2 processors, 4 GB RAM, and the Ubuntu 20.04 operating system, augmented by Raspberry Pi 4B devices serving as fog nodes and vehicle terminals. The maximum block generation interval is 2 seconds, with a bandwidth of 50-100 Mbps and a network latency of 10-200 ms. Hyperledger Fabric is one of the most widely used platforms for federation chains, along with enhanced flexibility. These features make it particularly suitable for high-throughput environments, such as parking lots. We configured 3 Orderer nodes and 5 Peer nodes. In Hyperledger, Orderer nodes are responsible for sorting transactions over time to ensure their correct order, while Peer nodes verify the correctness of transactions and store the blockchain ledger. Table III lists the default settings of the parameters.

TABLE III
Default Settings of Parameters

Parameters	Value
K	4
W_δ	0.5
W_ε	0.5
ϕ	$2^i\ (i = 1, 3, 5, ..)$
N	1200

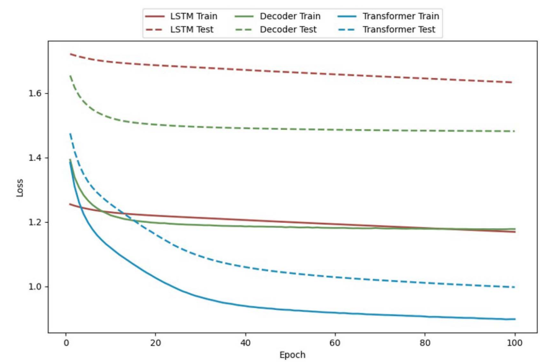

Fig. 8. Model Loss Function: The Transformer model demonstrates superior accuracy on both training and test sets. In convergence speed, it outperforms the Transformer-decoder, with LSTM exhibiting the slowest convergence.

B. Predicting Impact of Blocks

In this section, we complete the following experiment.

1) LSTM, Transformer-Decoder, and Transformer-Based Vehicle Trajectory and Short-Term Transaction Delivery Rate Prediction: The dataset comprises 18,932 vehicle behavior entries in (timestamp, longitude, latitude) format, representing 136 days of trip data. We classify data by date and normalize values to compute trajectory vectors, converting time intervals to hours to align magnitude with coordinate changes (e.g., (120, 0.023, -0.012) $\rightarrow$ (0.033, 0.023, -0.012)).

The blockchain dataset uses the structure (transaction arrival rate, block size, latency, throughput), with arrival rates ranging from 0 to 1000 and block sizes from 1 to 50. We measure corresponding latency and throughput through performance tests, then normalize arrival rates and block sizes to [0,1] via min-max scaling. For experiments, we configure a single-layer model with one attention head, batch size 1, and a 70:30 train-test split. Training employs mean squared error loss and stochastic gradient descent (learning rate=0.00001).

Fig. 8 reveals the Transformer achieves fastest convergence, followed by the Transformer-decoder, with LSTM being slowest. The Transformer demonstrates superior overall performance on both datasets. Analysis indicates the Transformer-decoder outperforms LSTM due to its attention mechanism capturing richer context, while the full Transformer excels by leveraging encoder-derived information from preceding frames to decode subsequent frames robustly.

2) Performance of Dynamic Blockchain: Different block sizes correspond to significantly varying scores in terms of

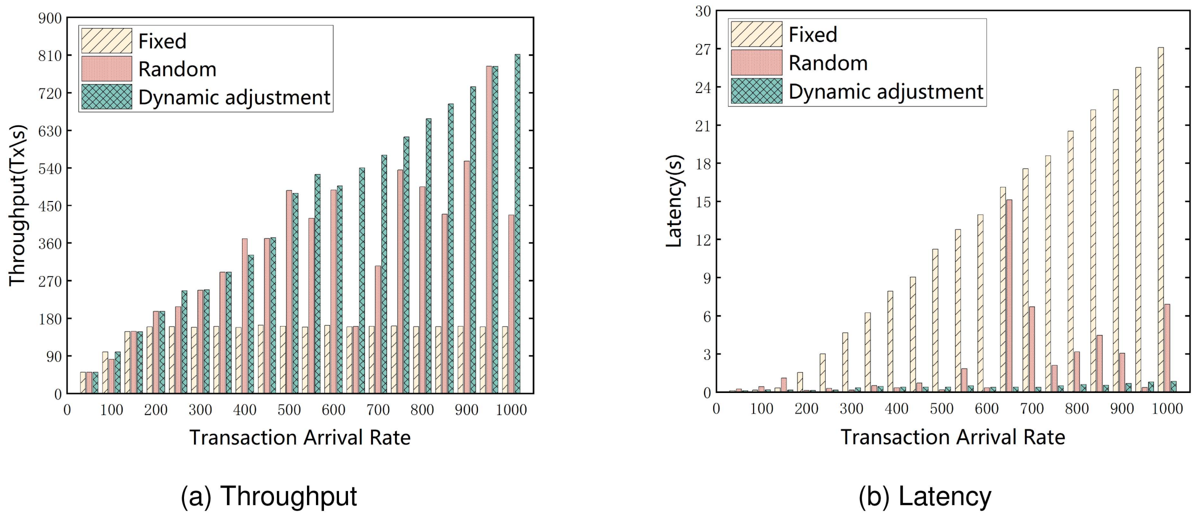

Fig. 9. Block performance and transaction arrival rate relationships: (a) Throughput; (b) Latency. Fixed block size (Bitcoin-standard 1 MB) exhibits constrained throughput and rapidly escalating latency. Random block size (1-50 MB) yields high variability in both metrics. We therefore dynamically optimize block size based on transaction arrival rate.

transaction arrival rates for different numbers of cross-domain vehicles. Selecting appropriate block sizes has a substantial impact on blockchain performance.

Fig. 9 evaluates blockchain performance under fixed (Bitcoin-standard 1 MB) versus random (1-50 MB) block sizes. Fixed sizing shows throughput plateauing at 150 Tx/s when transaction arrival rate reaches 150, while latency remains below 3 s initially but surges to 27.08 s beyond this threshold - indicating inadequacy for growing IoV demands. Random sizing causes throughput instability and latency spikes due to suboptimal block selection.

As shown in Fig. 9, the performance of the blockchain network is evaluated using our dynamic block resizing scheme. The overall throughput demonstrates a steady increase as the transaction arrival rate rises, starting at 49.5 Tx/s and reaching 811.6 Tx/s by the end, indicating a clear upward trend. Latency also shows an overall increasing trend but rises more gradually, starting at 0.13 s and ending at 0.85 s. Although the latency increases of 0.72 s is notable, it is significantly smaller compared to the fixed block size scheme (0.14 s to 27.08 s) and the random block size scheme (0.25 s to 15.12 s). Overall, our scheme dynamically adjusts the block size in real time based on the transaction arrival rate, offering superior flexibility and performance optimization.

As shown in Fig. 10, we use the Raspberry Pi 4B as the edge node to build the system platform. Its hardware configuration includes a 1.5 GHz quad-core processor and 8 GB of memory, with the operating system uniformly set to Ubuntu 24.10 (64-bit). The network bandwidth configuration is 50 Mbps upstream and 100 Mbps downstream. The experiment includes three scenario configurations:

- *Case 1:* Deploying 4 fog nodes with inter-node communication latency of 10–30 ms.
- *Case 2:* Deploying 8 fog nodes with inter-node communication latency of 50–100 ms.
- *Case 3:* Deploying 12 fog nodes with inter-node communication latency of 100–200 ms.

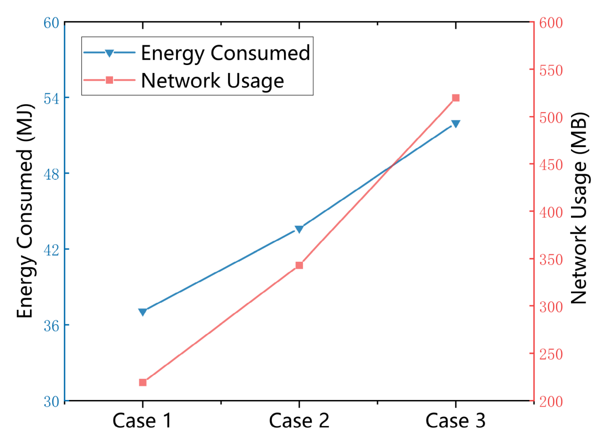

Fig. 10. Energy consumed and network usage.

Experimental results demonstrate that increased fog node count and communication latency elevate system energy consumption and network resource utilization. This stems from added operational overhead for node maintenance and high-latency communication processing. Nevertheless, analysis confirms this overhead growth remains within acceptable bounds.

C. Reputation-Authentication System Analysis

In this section, we demonstrate the role of reputation management in the cross-domain sharing scheme.

The experimental platform used is OMNET++. We configure a total of 4 RSU management domains. The probability of malicious behavior among vehicles is set to 10% and 30%, respectively, with a threshold value of 8. We conduct simulation experiments for the following two scenarios:

1) Single-Indicator Effects and Multiple-Indicator Effects: As shown in Fig. 11, a vehicle's reputation fluctuates minimally (1-2) under a single-indicator system, indicating limited deterrent effectiveness. Conversely, a multi-indicator system

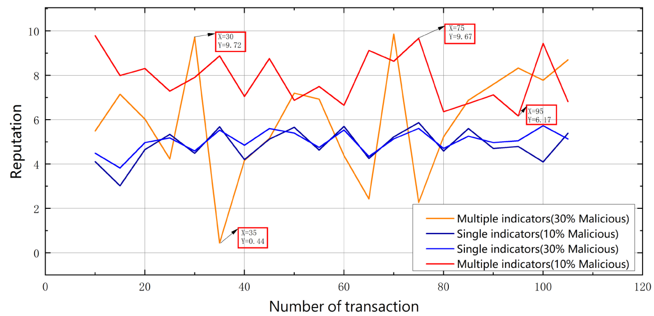

Fig. 11. Multi-indicator systems impact vehicle reputation more significantly than single-indicator approaches when malicious behavior occurs, enabling quicker and more effective constraints.

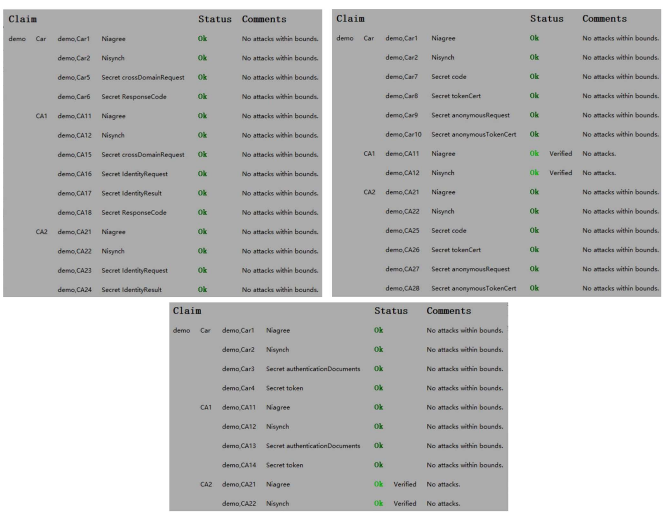

Fig. 12. Simulation results of the Scyther tool.

shows significant reputation impact: peak gaps reach 3 at 10% malicious behavior probability and 9 at 30%. Vehicles falling below the reputation threshold are barred from cross-domain sharing.

2) The Effect of Cross-Domain Authentication and Cross-Domain Authentication With Reputation: As shown in Fig. 12, we use the Scyther tool to perform formal verification on the proposed identity authentication protocol. The protocol is modeled as three stages: intra-domain authentication, cross-domain authorization code acquisition, and cross-domain authorization. The verification results show that the proposed protocol can pass Scyther's formal analysis, proving that it meets security requirements and is suitable for cross-domain IoV frameworks.

Fig. 13 shows multi-indicator region experiments with increasing vehicle counts over 500 s (30% malicious probability). Using only cross-domain authentication, transaction submission speed rises to 73 ms as vehicles approach 1,000 - reflecting increased authentication load. With reputation-augmented authentication, submission speed improves significantly: starting at 14 ms then dropping to 7 ms. Reputation blocks sub-threshold vehicles from transactions, eliminating malicious actors' authentication overhead and accelerating processing.

D. Performance of LMIRA

Fig. 14 compares the throughput, latency, and number of transactions for the PAA [36], Wang [37], BEET [38], BTSR [39], and LMIRA frameworks. It is evident that our framework significantly outperforms the PAA, Wang, BEET, and BTSR frameworks in terms of both throughput and latency. As shown in Fig. 14(a), when the number of transactions is below 2000, the throughput of our framework does not significantly surpass that of the PAA, Wang, BEET, and BTSR frameworks. This is because our block size is optimized for lower latency at lower transaction volumes. However, as the number of transactions increases, the throughput of the other frameworks stabilizes within a certain range, while our framework continues to exhibit increasing throughput, demonstrating its adaptability to varying transaction scenarios.

Fig. 14(b) illustrates that our framework exhibits significantly lower latency compared to PAA, Wang, BEET, and BTSR. At 2000 transactions, our latency stands at 0.15 seconds, whereas the latencies of other methods exceed 0.9 seconds. As the transaction volume increases and approaches the processing limit, latency spikes rapidly. BEET peaks at 6000 transactions, PAA peaks at 9000 transactions, and BTSR peaks at 2000 transactions, followed by a sudden drop and slow recovery. Meanwhile, Wang experiences severe performance degradation before reaching the official threshold. These advantages stem from our dynamic block size adjustment and efficient framework implementation.

Fig. 15 compares failure rates versus transaction volume for PAA, Wang, BEET, BTSR and our LMIRA framework under high-stress simulation. As transaction requests increase, failure rates universally rise. LMIRA maintains stable failure rates between 10% and 30%, measuring 7.8% at 2,000 transactions and 31.18% at 20,000 transactions. Competitors exhibit significantly higher failures, rising from initial 20-30% to 70-100% at peak loads, indicating system breakdown requiring full resource recovery.

In contrast, the LMIRA system predicts a more suitable block size when 16,000 transaction requests are submitted, owing to its dynamic block size adjustment mechanism. As a result, the transaction submission failure rate decreases by 6.41% after 14,000 transactions. This comparative demonstrates the LMIRA framework's sustained superior system stability.

The BEET, Wang, PAA and BTSR frameworks all demonstrate promising performance in the vehicular network domain, leveraging distinct algorithmic advantages. Nevertheless, their aggregate performance falls slightly short of that of the LMIRA framework. Regarding blockchain load, query operations are inherently read-only and do not necessitate new block generation, consequently exerting minimal load impact. However, practical operation involves continuous read and write operations due to the frequent processing of substantial shared data. Although meticulously designed smart contracts can enhance write operation throughput and reduce latency to a certain degree, they fail to address the fundamental scalability limitations inherent to blockchain technology. For smaller blocks, irrespective of smart contract optimizations, the system rapidly encounters a

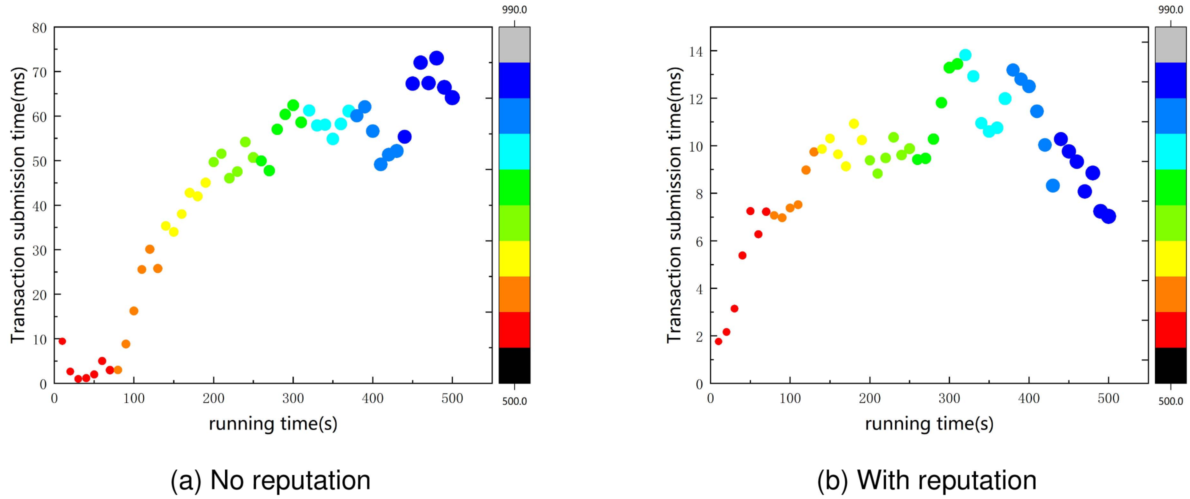

Fig. 13. Impact of credibility mechanisms on the submission of information. (a) Without reputation; (b) With reputation (coordinate color = vehicle density). Reputation mechanisms effectively constrain malicious submissions and enhance transaction efficiency, yielding significant acceleration in transaction sharing speeds.

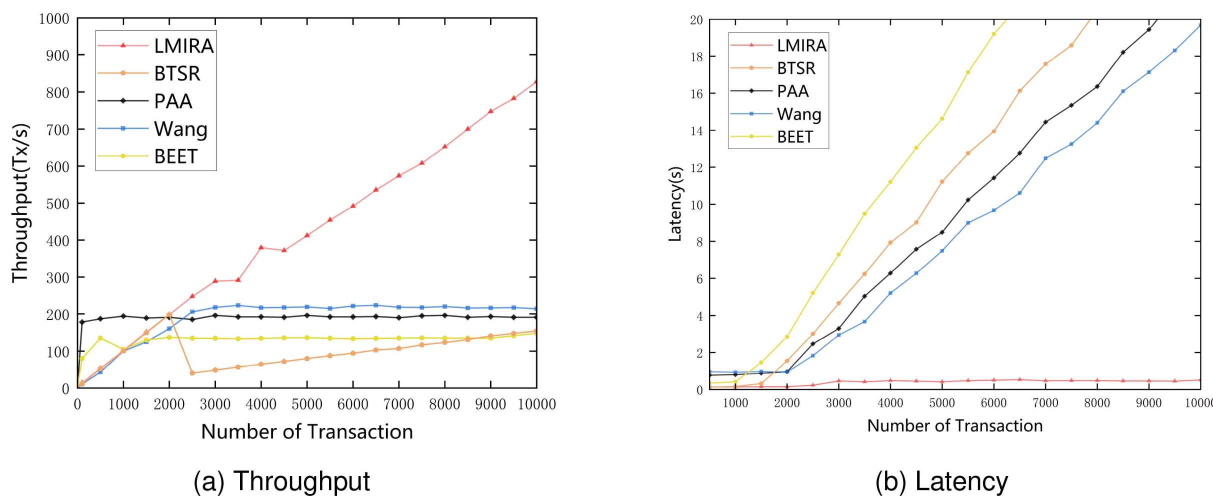

Fig. 14. Throughput and latency comparison of different frameworks. (a) Throughput comparison. (b) Latency comparison. The LMIRA framework maintains high throughput as transaction volume increases, with only a marginal rise in latency.

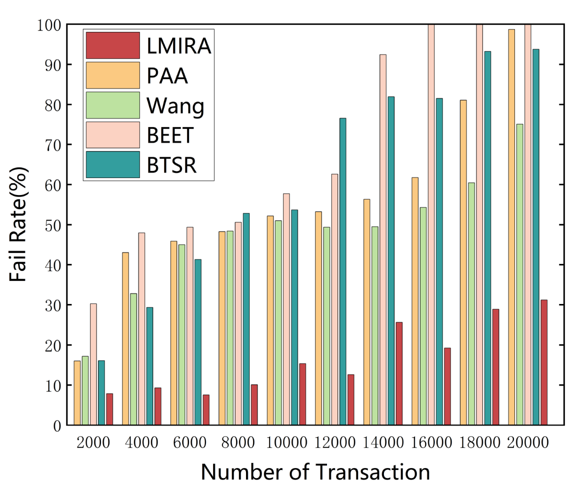

Fig. 15. Fail rate comparison of different frameworks. The LMIRA framework exhibits a gradual increase in transaction submission failure rate under high stress and remains operational even when other frameworks fail.

throughput bottleneck, leading to constrained throughput and significantly elevated latency. For larger blocks, generating them demands aggregating a larger number of transactions. During periods of low transaction volume, this aggregation requirement hinders meeting the minimum block generation threshold. Additionally, larger block sizes incur higher network bandwidth consumption, collectively resulting in diminished throughput and increased latency.

The LMIRA dynamically adapts block size based on real-time network traffic conditions. This intelligent adaptive mechanism ensures the continuous adoption of an optimal block size, thereby maximizing throughput and minimizing latency.

VIII. Conclusion

This paper proposes LMIRA, a load-balanced multi-indicator reputation and authentication blockchain framework for cross-domain vehicular data sharing. LMIRA employs edge computing to deploy a Transformer-based predictor forecasting transaction arrival rates and block sizes for RSU-managed vehicles. It further establishes a cross-domain reputation sharing and authentication architecture. Extensive experiments validate the framework's feasibility and innovation.

This study has several limitations. The Transformer-based prediction depends heavily on historical data quality and may underperform in highly dynamic traffic. The multi-indicator reputation system adds computational overhead and requires environment-specific tuning. The cross-domain authentication scheme assumes fully trustworthy CAs, which may not hold in practice. Finally, the dynamic block-size adjustment, while improving latency and throughput, is still constrained by consensus protocol overhead and network bandwidth.

Future work will develop advanced trajectory and block-size prediction models to boost performance and strengthen data security. Insights from AI-adaptive learning systems [40] may inform more intelligent IoV designs. Specific directions include: robust prediction for sudden traffic changes, RL-based reputation weight adaptation, decentralized trust to reduce CA reliance, and lightweight consensus for lower latency and energy in resource-constrained environments.

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